

Question Bank - Basic [SQL]

Consider a example table as -

- step 1 - shoot following query to create a table-

```
1 CREATE TABLE student (  
2     id INT AUTO_INCREMENT PRIMARY KEY,  
3     name VARCHAR(50),  
4     age INT,  
5     email VARCHAR(100),  
6     gender VARCHAR(10),  
7     grade CHAR(1),  
8     marks INT,  
9     city VARCHAR(50),  
10    registration_date DATE  
11 );  
12
```

- step 2 - Shoot following query to insert data in table

```
1 INSERT INTO student (name, age, email, gender, grade, marks, city, registration_date) VALUES  
2 ('Alice', 20, 'alice@example.com', 'Female', 'A', 88, 'Delhi', '2024-06-01'),  
3 ('Bob', 21, 'bob@example.com', 'Male', 'B', 75, 'Mumbai', '2024-05-21'),  
4 ('Carol', 22, 'carol@example.com', 'Female', 'C', 64, 'Chennai', '2024-03-14'),  
5 ('David', 19, 'david@example.com', 'Male', 'A', 92, 'Pune', '2024-04-10'),  
6 ('Eva', 23, 'eva@example.com', 'Female', 'B', 78, 'Delhi', '2024-06-10'),  
7 ('Frank', 20, 'frank@example.com', 'Male', 'C', 55, 'Bangalore', '2024-01-29'),  
8 ('Grace', 18, 'grace@example.com', 'Female', 'A', 95, 'Mumbai', '2024-05-05'),  
9 ('Harry', 21, 'harry@example.com', 'Male', 'B', 70, 'Kolkata', '2024-06-01'),  
10 ('Ivy', 20, 'ivy@example.com', 'Female', 'C', 60, 'Hyderabad', '2024-02-15'),  
11 ('Jack', 22, 'jack@example.com', 'Male', 'A', 85, 'Delhi', '2024-03-20'),  
12 ('Kathy', 24, 'kathy@example.com', 'Female', 'B', 77, 'Chennai', '2024-04-25'),  
13 ('Liam', 19, 'liam@example.com', 'Male', 'C', 58, 'Pune', '2024-01-01'),  
14 ('Mona', 21, 'mona@example.com', 'Female', 'A', 90, 'Delhi', '2024-05-30'),  
15 ('Nate', 20, 'nate@example.com', 'Male', 'B', 74, 'Bangalore', '2024-03-15'),  
16 ('Olivia', 22, 'olivia@example.com', 'Female', 'C', 69, 'Kolkata', '2024-04-12'),  
17 ('Paul', 23, 'paul@example.com', 'Male', 'A', 83, 'Mumbai', '2024-06-18'),  
18 ('Quinn', 18, 'quinn@example.com', 'Female', 'B', 72, 'Hyderabad', '2024-05-22'),  
19 ('Rita', 20, NULL, 'Female', 'C', 65, 'Chennai', '2024-03-01'),  
20 ('Steve', 21, 'steve@example.com', 'Male', 'A', 88, 'Delhi', '2024-06-24'),  
21 ('Tina', 19, 'tina@example.com', 'Female', 'B', 76, 'Pune', '2024-04-02');  
22
```

✓ Basic SQL Questions (Single Table)

♦ SELECT & FILTERING (Q1-Q15)

1. Select all columns from the student table.
2. Select only names and ages of students.
3. Find students who are older than 18.
4. Find students who are exactly 20 years old.
5. Find students with marks greater than or equal to 75.
6. Find students living in 'Delhi'.

7. **Find female students.**
 8. **Find students whose name starts with 'A'.**
 9. **Find students whose name ends with 'n'.**
 10. **Find students whose name contains 'an'.**
 11. **Find students whose marks are between 60 and 80.**
 12. **Find students whose city is either 'Delhi', 'Mumbai', or 'Chennai'.**
 13. **Find students who are not from 'Pune'.**
 14. **Find students who are not female.**
 15. **Find students with null emails.**
-

♦ ORDER BY, LIMIT (Q16–Q20)

16. **List all students sorted by marks in descending order.**
 17. **List top 5 students with highest marks.**
 18. **List bottom 3 students with lowest marks.**
 19. **List students ordered by name alphabetically.**
 20. **List students registered most recently (latest date first).**
-

♦ AGGREGATE FUNCTIONS (Q21–Q30)

21. **Count total number of students.**
 22. **Find the average age of students.**
 23. **Find the maximum marks.**
 24. **Find the minimum age.**
 25. **Find the total marks of all students.**
 26. **Count how many students are from 'Delhi'.**
 27. **Count male and female students.**
 28. **Find average marks for each grade.**
 29. **Count number of students in each city.**
 30. **Find max, min, and avg marks.**
-

♦ GROUP BY & HAVING (Q31–Q35)

31. **Group students by age.**
32. **Find cities with more than 3 students.**
33. **Find grades with average marks above 80.**

34. List all genders with at least 2 students.

35. Find ages with more than 1 student.

♦ MODIFY TABLE STRUCTURE (Q36–Q40)

36. Add a new column `phone` to the student table.

37. Change the `grade` column from `CHAR(1)` to `VARCHAR(2)`.

38. Rename the `marks` column to `score`.

39. Drop the column `email`.

40. Set default value for city as 'Unknown'.

♦ UPDATE & DELETE (Q41–Q45)

41. Update marks to 100 for student named 'John'.

42. Increase all student marks by 5.

43. Set city to 'Mumbai' for all students with null city.

44. Delete all students with marks less than 35.

45. Delete students from 'Chennai'.

♦ MISCELLANEOUS (Q46–Q50)

46. Get current date in SQL.

47. Find students registered in year 2024.

48. Find length of each student's name.

49. Convert student names to uppercase.

50. Display names with 'Mr.' prefix (use CONCAT).
